

Practical Demo: Duration, Convexity, and Portfolio Repricing

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This reading copy shows the calculations' results and figures. The HTML demo contains the full code in folded blocks and interactive scenario controls.

The desk question

A manager holds three cash bonds and asks: **if yields rise by 50 basis points, how much will we lose, and can a duration estimate be trusted?** Start with the payments you own, then measure risk. Do not start by averaging the bonds' yields.

1. **Read the position file:** distinguish face amount from market value.
2. **Discount each payment:** recover the price and the timing of its present value.
3. **Measure sensitivity:** translate duration into dollars per basis point.
4. **Shock and reprice:** compare a tangent-line estimate with actual discounted cash flows.
5. **Report the exception:** explain why a curve twist needs more than one risk number.

Read the results first; open **Show calculation** when you want to reproduce them. The interactive controls work without a Python installation; changing the source inputs requires rerunning the notebook from the top.

Purpose and Assumptions

This demo prices a small bond portfolio from contractual cash flows, measures Macaulay duration, modified duration, DV01/PVBP, and convexity, and compares approximations with exact repricing. All bonds and yields are **illustrative, market-like inputs**, not live quotes. Bonds pay semiannual coupons; yields are nominal annual yields compounded semiannually; settlement is on a coupon date; default, taxes, accrued interest, and embedded options are ignored.

Workflow stage	In this case
Observed on a position report	Face amount, coupon, maturity, and yield
Calculated	Full price, duration, DV01, and convexity
Checked	Approximation error against exact cash-flow repricing
Used for	Portfolio shock P&L and identification of curve risk

1 Full Cash-Flow Pricing and Risk Measures

For cash flow CF_k at semiannual period k ,

$$P = \sum_{k=1}^n \frac{CF_k}{(1 + y/m)^k}.$$

Macaulay duration is the present-value-weighted average payment time. Modified duration is $D_{Mac}/(1 + y/m)$, and

$$DV01 = PD_{mod}(0.0001).$$

Bond	2-year Treasury	5-year agency	10-year corporate
Face value (\$mm)	12	10	8
Coupon rate	0.04	0.0475	0.0575
Maturity (years)	2	5	10
Yield	0.041	0.0485	0.061
Price (\$)	11,977,181.3216	9,956,069.5998	7,792,674.9472
Macaulay duration	1.9419	4.5077	7.7068
Modified duration	1.9029	4.401	7.4787
DV01/PVBP (\$/bp)	2,279.1012	4,381.6687	5,827.9378
Convexity	4.6154	22.7992	69.2981

2 Exact Repricing Versus Approximations

i Follow one payment

The corporate bond pays \$230,000 every six months: $\$8,000,000 \times 5.75\% / 2$. Its first payment is worth $\$230,000 / 1.0305$ today, while its final payment is discounted twenty times. Later payments are more sensitive because a yield change affects more discounting periods. This is why maturity alone does not describe the portfolio's interest-rate risk.

For a yield shock Δy ,

$$\frac{\Delta P}{P} \approx -D_{mod}\Delta y$$

and the duration-plus-convexity approximation is

$$\frac{\Delta P}{P} \approx -D_{mod}\Delta y + \frac{1}{2}C(\Delta y)^2.$$

	Yield shock (bp)	Duration-only change (%)	Duration + convexity change (%)
0	-150	11.218	11.998
1	-75	5.609	5.804
2	-25	1.87	1.891
3	25	-1.87	-1.848
4	75	-5.609	-5.414

	Yield shock (bp)	Duration-only change (%)	Duration + convexity change (%)
5	150	-11.218	-10.439

	Exact price change (%)	Duration-only error (bp of price)	Duration + convexity error (bp of price)
0	12.039	-82.092	-4.132
1	5.809	-19.996	-0.506
2	1.892	-2.184	-0.018
3	-1.848	-2.147	0.018
4	-5.419	-19.004	0.486
5	-10.477	-74.151	3.809

2.1 Explore the shock size (HTML)

The table and figure above are the static fallback in PDF. In HTML, move the slider to reprice the same 10-year 5.75% coupon bond exactly.

3 Portfolio Aggregation

Dollar risk aggregates across positions. Portfolio DV01 is the sum of bond DV01s; market-value-weighted duration is equivalent to total dollar duration divided by portfolio value.

```
Portfolio market value ($)      2.9726e+07
Portfolio DV01/PVBP ($/bp)    1.2489e+04
Portfolio modified duration     4.2010e+00
Market-value-weighted convexity 2.7662e+01
dtype: float64
```

4 Parallel and Nonparallel Scenarios

A single duration number is designed for a common yield shift. Here we approximate a curve twist with different yield-to-maturity shocks for the three bonds. Repricing is exact for those

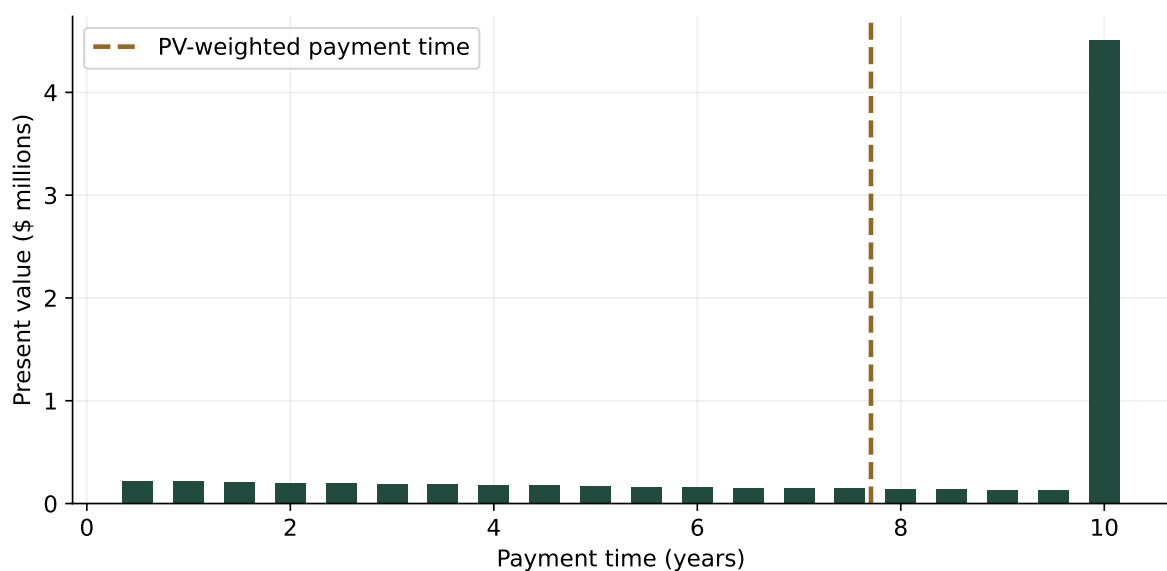


Figure 1: Present value by payment date for the corporate position; the final bar includes principal.

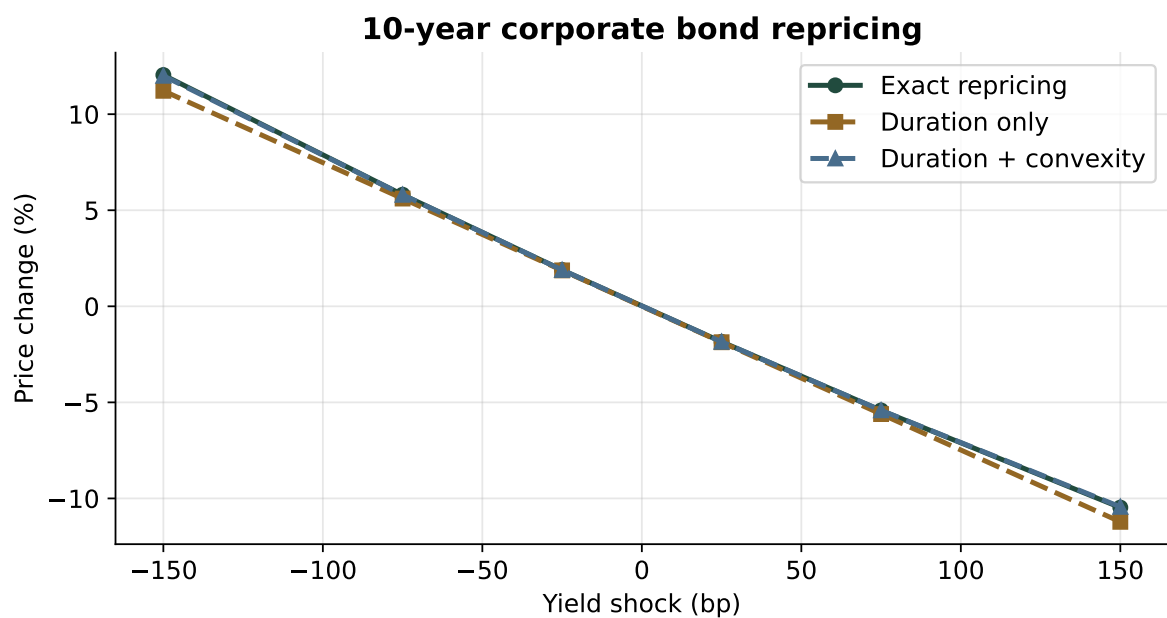


Figure 2: Convexity materially improves the approximation for larger yield moves

assigned bond yields, not for a fully specified spot-curve twist. A production key-rate analysis would shock the underlying discount curve and revalue every dated cash flow.

Scenario	Duration P&L (\$)	Duration + convexity P&L (\$)	Exact P&L (\$)
Parallel +50 bp	-624,435	-614,157	-614,300
Bear steepener	-613,245	-596,639	-597,027
Bull flattener	755,876	780,107	780,811

The parallel scenario can be summarized by portfolio duration. The steepener and flattener require maturity-specific exposures; the same portfolio DV01 can hide materially different curve risk.

5 Validation Checks

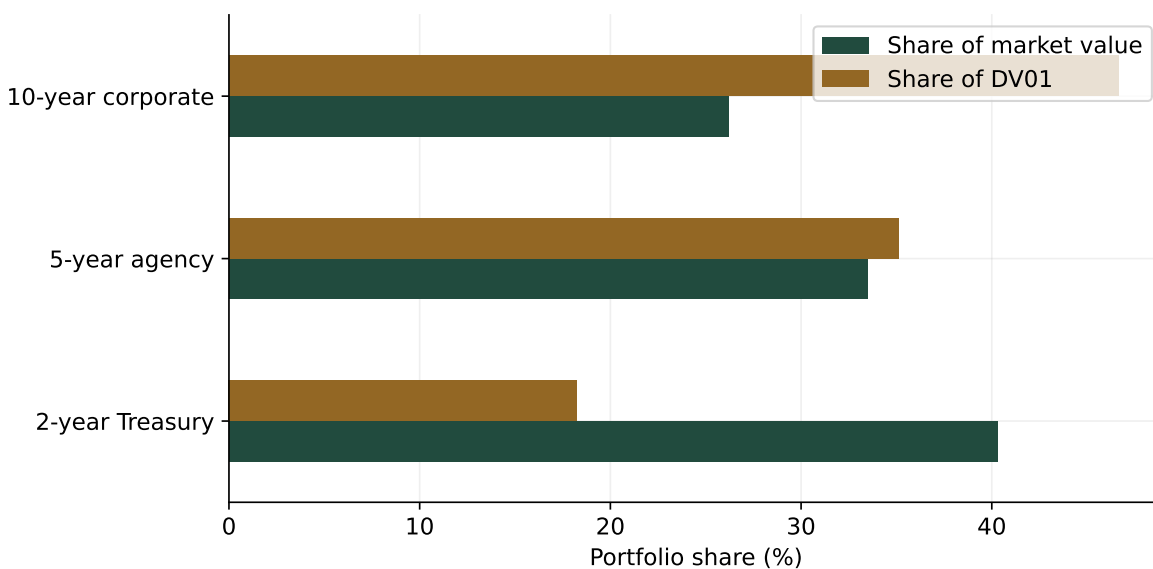


Figure 3: Capital allocation and risk allocation answer different questions.

💡 Predict, change, explain

Move the yield-shock slider from +25 to +150 bp, then to -150 bp. Are equal upward and downward shocks symmetric in price? Compare the exact result with the duration-only table. Next, identify the position with the largest share of DV01: selling the largest face amount is not necessarily the most efficient risk reduction.

```
Macaulay duration exceeds modified duration    True
DV01 values are positive                      True
Portfolio DV01 equals component sum           True
Convexity improves average approximation      True
Exact price falls as yield rises              True
dtype: bool
```

6 Financial Interpretation

Duration is a local slope, convexity is a curvature correction, and exact cash-flow repricing is the benchmark. DV01 translates percentage sensitivity into dollars. Portfolio aggregation is reliable for a stated parallel shock, but nonparallel scenarios require maturity-specific or key-rate exposures.

6.1 From the classroom to a position report

Use [FINRA Fixed Income Data](#) to inspect security terms and recent trades. Record the trade date, settlement date, coupon, maturity, price type, and accrued interest before substituting a bond. TRACE records trades, not a guaranteed executable quote. The embedded position table remains the complete offline case; its numbers are not FINRA observations.

Your deliverable: a short risk note stating portfolio value, DV01, the exact +50 bp loss, approximation error in dollars, and one nonparallel scenario.

7 Guided explorer

Start at a zero shock, then compare +150 and -150 bp. The straight-line estimate has no curvature. Explain which approximation is closest and why equal yield moves do not produce equal percentage gains and losses.

The HTML control selects a precomputed scenario. In the PDF, use the preceding figures and tables and rerun the displayed notebook inputs to explore other values.